

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	<i>Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.</i>		
Student Name:	Siva Charan	Reg. No.:	192525039
Section / Batch:	AI&ML	Date:	31/08/2026

Code of Conduct

I Siva Charan (192525039)_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this
assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Siva Charan_____

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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QUESTIONS

Q1 Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2 Graph - Topological Sort
20 marks

Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

Your Answer

ALGORITHM: Kruskal's Minimum Spanning Tree
INPUT: Graph G with vertices V (zones) and edges E (possible cable connections with installation costs)
OUTPUT: MST containing minimum cost connections linking all zones

BEGIN

// Step 1: Initialization

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- Detect cycle before sorting
- Use DFS approach

Your Answer

```
ALGORITHM TopologicalSortDFS(Graph G)
// Initialize state for each node
// States: 0 = UNVISITED, 1 = VISITING, 2 = VISITED
FOR each vertex v in G.Vertices:
    state[v] = UNVISITED
```

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- a. Use MST for infrastructure
- b. Use shortest path for routing

Your Answer

ALGORITHM SmartTransportation_MST_And_Dijkstra(Graph G, Source s)

INPUT:

$G = (V, E)$ where:

V = junctions/stations/locations

E = roads/edges with weights (cost/length)

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
ALGORITHM TollFreeDijkstra(G, S, D):  
    # G is a graph with weighted edges (u, v, w)  
    # Voucher allows taking exactly one edge at cost 0  
    # State = (node, usedVoucher) where usedVoucher ∈ {0, 1}  
  
    for each node i in G:
```

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

SECOND_BEST_MST(G):
Input: Weighted undirected connected graph $G=(V,E)$
Output: Weight of the second best MST (or INF if not exists)

Step 1: Find the MST using Kruskal (also works with Prim if you store parent edges)
MST = KRUSKAL_MST(G) # list of edges in MST

 **Save Answer**